

Question 3

```
disp('Explanation of Kuwahara Filter Process and Observations:');
```

Explanation of Kuwahara Filter Process and Observations:

```
disp('The Kuwahara filter, applied here with a 5x5 window, is designed to preserve edge features while reducing noise in homogeneous areas.');
```

The Kuwahara filter, applied here with a 5x5 window, is designed to preserve edge features while reducing

```
disp('This filter works by dividing the window into four overlapping sub-windows. Each sub-windows variance and mean are calculated.');
```

This filter works by dividing the window into four overlapping sub-windows. Each sub-windows variance and

```
disp('The output pixel is set to the mean of the sub-window with the lowest variance, ensuring that edge details are preserved while smoothing areas without strong edges.');
```

The output pixel is set to the mean of the sub-window with the lowest variance, ensuring that edge details

```
disp('This method is particularly effective in images with clear and smooth areas separated by sharp edges, making it ideal for images with significant texture variation.');
```

This method is particularly effective in images with clear and smooth areas separated by sharp edges, making

```
disp('The difference image highlights where the filter has made significant changes, typically around the edges and in regions with noise.');
```

The difference image highlights where the filter has made significant changes, typically around the edges

```
disp('Histogram comparisons provide a quantitative measure of the changes, showing how pixel intensities are distributed before and after filtering.');
```

Histogram comparisons provide a quantitative measure of the changes, showing how pixel intensities are distributed

```
disp('Such visual and numerical analysis helps in understanding the filter's impact on image aesthetics and quality.');
```

Such visual and numerical analysis helps in understanding the filter's impact on image aesthetics and quality

```
function output = kuwaharaFilter(img, ksize)
    if mod(ksize, 2) == 0
        error('Kernel size must be odd.');
```

```
    end

    padSize = floor(ksize/2);
    paddedImg = padarray(img, [padSize padSize], 'replicate', 'both');

    [rows, cols] = size(img);
    output = zeros(rows, cols);
```

```

for i = 1:rows
    for j = 1:cols

        center_i = i + padSize;
        center_j = j + padSize;

        sub1 = paddedImg(center_i-padSize:center_i-1, center_j-
padSize:center_j-1);
        sub2 = paddedImg(center_i-padSize:center_i-1,
center_j+1:center_j+padSize);
        sub3 = paddedImg(center_i+1:center_i+padSize, center_j-
padSize:center_j-1);
        sub4 = paddedImg(center_i+1:center_i+padSize,
center_j+1:center_j+padSize);

        means = [mean(sub1(:)), mean(sub2(:)), mean(sub3(:)),
mean(sub4(:))];
        vars = [var(double(sub1(:))), var(double(sub2(:))),
var(double(sub3(:))), var(double(sub4(:)))]];

        [~, idx] = min(vars);

        output(i, j) = means(idx);
    end
end

output = uint8(255 * mat2gray(output));
end

img = imread('Bat.jpg');
gray_img = rgb2gray(img);
filtered_img = kuwaharaFilter(gray_img, 5);
figure;
subplot(1,2,1), imshow(gray_img), title('Original Grayscale Image');
subplot(1,2,2), imshow(filtered_img), title('Kuwahara Filtered Image');

```

Original Grayscale Image



Kuwahara Filtered Image



```
% Load the image one more time to check with the difference with kuwahara
img = imread('Bat.jpg');
gray_img = rgb2gray(img);
filtered_img = kuwaharaFilter(gray_img, 5);
figure;
subplot(1,3,1), imshow(gray_img), title('Original Grayscale Image');
subplot(1,3,2), imshow(filtered_img), title('Kuwahara Filtered Image');
diff_img = abs(double(gray_img) - double(filtered_img));
subplot(1,3,3), imshow(diff_img, []), title('Difference Image');
```

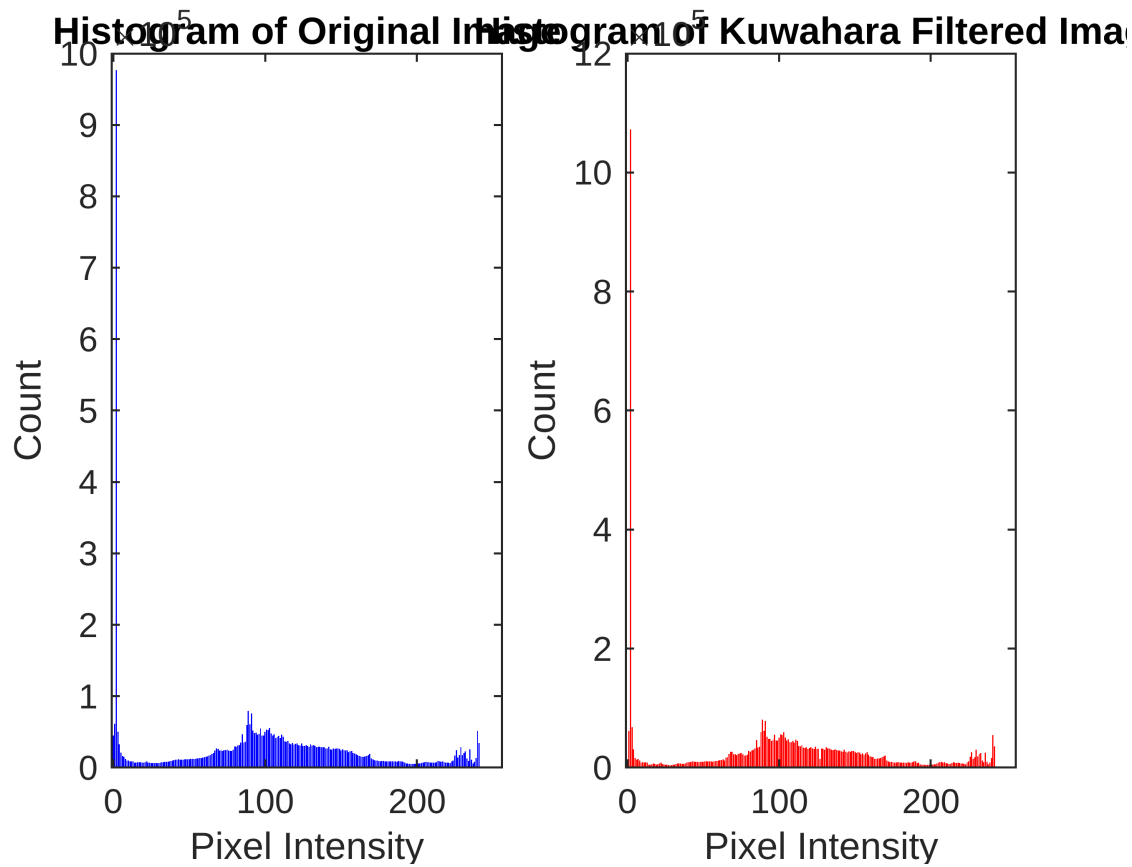
Original Grayscale Image Kuwahara Filtered Image Difference Image



```
[hist_orig, orig_bins] = imhist(gray_img);
[hist_filt, filt_bins] = imhist(filtered_img);

figure;
subplot(1,2,1);
bar(orig_bins, hist_orig, 'b');
title('Histogram of Original Image');
xlabel('Pixel Intensity'); ylabel('Count');

subplot(1,2,2);
bar(filt_bins, hist_filt, 'r');
title('Histogram of Kuwahara Filtered Image');
xlabel('Pixel Intensity'); ylabel('Count');
```



```
disp('Histogram Values for Original Image:');
```

Histogram Values for Original Image:

```
disp([orig_bins, hist_orig]);
```

0	45146
1	61503
2	976939
3	50033
4	32732
5	21179
6	16502
7	15117
8	12287
9	9819
10	9822
11	8988
12	9001
13	9019
14	6758
15	6805
16	7482
17	7603
18	7402
19	6870
20	6884
21	7630
22	8525
23	7241
24	6858

25	6505
26	6552
27	6401
28	6320
29	6348
30	6711
31	7080
32	7623
33	7870
34	7970
35	8058
36	8298
37	8764
38	9203
39	10152
40	10565
41	10844
42	11281
43	11665
44	11384
45	11087
46	11223
47	11489
48	11514
49	11861
50	11886
51	12094
52	12152
53	12372
54	12474
55	12686
56	12816
57	13411
58	13663
59	14051
60	14590
61	15407
62	15598
63	16898
64	17497
65	19316
66	20278
67	24182
68	26728
69	26372
70	24021
71	24076
72	23150
73	24827
74	24321
75	25308
76	24122
77	23427
78	23411
79	25135
80	29946
81	29067
82	31201
83	31816
84	34876
85	46967
86	35317
87	36214
88	59680

89	79699
90	60626
91	75837
92	51946
93	48654
94	49158
95	45979
96	47167
97	54715
98	45193
99	44995
100	50013
101	53282
102	52512
103	55564
104	48016
105	44850
106	46541
107	41489
108	43041
109	44533
110	41353
111	45978
112	43077
113	37016
114	36056
115	37460
116	33757
117	33002
118	34288
119	32452
120	33103
121	33839
122	31487
123	31047
124	33742
125	30544
126	30628
127	31493
128	30365
129	29515
130	32772
131	30822
132	31646
133	30263
134	28678
135	28980
136	29439
137	28693
138	28750
139	28623
140	26662
141	26970
142	29471
143	25877
144	25406
145	26740
146	26048
147	27164
148	27016
149	26119
150	23989
151	25523
152	24340

153	23833
154	24497
155	21858
156	22915
157	23459
158	20437
159	19636
160	18520
161	17848
162	16435
163	15797
164	15435
165	15305
166	15609
167	16424
168	17545
169	19503
170	12880
171	11540
172	10788
173	10161
174	9811
175	9296
176	9088
177	9202
178	9270
179	8991
180	8816
181	8639
182	8677
183	8566
184	8792
185	8794
186	8526
187	8430
188	9319
189	8953
190	8782
191	7943
192	6967
193	6032
194	5886
195	5415
196	5203
197	5292
198	5270
199	5365
200	5476
201	5617
202	6128
203	6364
204	7165
205	7663
206	8368
207	7708
208	7908
209	7281
210	7195
211	7055
212	7068
213	7646
214	9601
215	8701
216	8143

217	8683
218	7837
219	7162
220	7034
221	6857
222	6269
223	8541
224	10028
225	17196
226	24605
227	13802
228	17355
229	28629
230	17394
231	20361
232	22816
233	12701
234	9623
235	25637
236	10867
237	6125
238	8124
239	13672
240	51180
241	34488
242	1593
243	880
244	478
245	254
246	152
247	104
248	66
249	45
250	31
251	17
252	24
253	19
254	16
255	1

```
disp('Histogram Values for Kuwahara Filtered Image:');
```

Histogram Values for Kuwahara Filtered Image:

```
disp([filt_bins, hist_filt]);
```

0	9691
1	61515
2	1072473
3	67776
4	30883
5	16018
6	13058
7	14793
8	11876
9	8649
10	9001
11	8492
12	9112
13	8279
14	4717
15	4760
16	5903

17	6871
18	6340
19	5181
20	5492
21	6948
22	8369
23	6386
24	5352
25	4940
26	4843
27	4569
28	3955
29	4301
30	4600
31	5085
32	6321
33	6939
34	6817
35	6895
36	6484
37	6523
38	7308
39	8508
40	8996
41	9048
42	9976
43	10210
44	9815
45	9527
46	9681
47	9243
48	9518
49	9676
50	9890
51	10397
52	10509
53	10433
54	10369
55	10755
56	10137
57	10592
58	11420
59	11344
60	11708
61	12885
62	12532
63	14111
64	11678
65	16786
66	17911
67	23245
68	26870
69	26384
70	22546
71	22277
72	21249
73	23345
74	23676
75	24545
76	22500
77	20691
78	20575
79	21836
80	27826

81	26928
82	28932
83	30000
84	32636
85	46127
86	33904
87	34905
88	59300
89	80406
90	61397
91	78624
92	52105
93	48299
94	48230
95	45085
96	46827
97	55395
98	45517
99	45480
100	50655
101	55773
102	54416
103	59578
104	50217
105	45658
106	48006
107	42559
108	42951
109	44612
110	41813
111	47075
112	45004
113	36479
114	36127
115	37549
116	33814
117	33055
118	34698
119	31331
120	32957
121	34279
122	32323
123	31826
124	35210
125	31615
126	31294
127	15011
128	31951
129	31703
130	30494
131	34286
132	32377
133	32380
134	31125
135	29640
136	29181
137	30802
138	29217
139	28984
140	28494
141	27210
142	26802
143	30256
144	26495

145	25168
146	27066
147	26681
148	28108
149	28254
150	26641
151	24380
152	25884
153	25016
154	22749
155	23737
156	21219
157	24177
158	26153
159	21848
160	18748
161	18448
162	17506
163	14949
164	14595
165	15208
166	15346
167	16060
168	17624
169	19012
170	20066
171	12129
172	10468
173	9480
174	10045
175	8833
176	8756
177	8602
178	9002
179	9278
180	8803
181	8686
182	8249
183	8750
184	8091
185	9192
186	8689
187	8432
188	9214
189	10405
190	10337
191	7813
192	8238
193	5950
194	5093
195	5161
196	4729
197	4553
198	4735
199	4817
200	4724
201	5252
202	5404
203	5938
204	6878
205	8488
206	9187
207	10065
208	8686

209	8857
210	7497
211	7320
212	6684
213	6616
214	7616
215	9471
216	8793
217	7971
218	8418
219	7485
220	6724
221	6916
222	6654
223	5824
224	8947
225	10738
226	18300
227	26100
228	14616
229	17698
230	30414
231	18962
232	23085
233	24726
234	12285
235	9178
236	25361
237	10172
238	5869
239	8754
240	16014
241	54591
242	36001
243	716
244	175
245	128
246	62
247	29
248	13
249	14
250	15
251	7
252	15
253	10
254	6
255	16